**Grade 11 – Final Exam Syllabus Focus – MAY 2026**

| SUBJECT | PORTION | COMPONENT AND MARKS |
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| **Biology** | **Unit 1: MOLECULES – 60 hours**<br><br>**A1.1 Water**<br>• Introduction to biochemistry/molecular biology<br>A1.1.1—Water as the medium for life<br>A1.1.2—Hydrogen bonds as a consequence of the polar covalent bonds within water molecules<br>A1.1.3—Cohesion of water molecules due to hydrogen bonding and consequences for organisms<br>A1.1.4—Adhesion of water to materials that are polar or charged and impacts for organisms<br>A1.1.5—Solvent properties of water linked to its role as a medium for metabolism and for transport in plants and animals<br>A1.1.6—Physical properties of water and the consequences for animals in aquatic habitats<br>A1.1.7—Extraplanetary origin of water on Earth and reasons for its retention<br>A1.1.8—Relationship between the search for extraterrestrial life and the presence of water<br><br>**B1.1 Carbohydrates and lipids**<br>B1.1.1—Chemical properties of a carbon atom allowing for the formation of diverse compounds upon which life is based<br>B1.1.2—Production of macromolecules by condensation reactions that link monomers to form a polymer<br>B1.1.3—Digestion of polymers into monomers by hydrolysis reactions<br>B1.1.4—Form and function of monosaccharides<br>B1.1.5—Polysaccharides as energy storage compounds<br>B1.1.6—Structure of cellulose related to its function as a structural polysaccharide in plants<br>B1.1.7—Role of glycoproteins in cell–cell recognition<br>B1.1.8—Hydrophobic properties of lipids | **SL Paper 1**<br>1 Hr 30 mins, 55 Marks<br><br>**SL Paper 2**<br>1 Hr 30 Mins, 50 Marks<br><br>**HL Paper 1**<br>2 Hrs, 75 Marks<br><br>**HL Paper 2**<br>2 Hrs 30 Mins, 80 Marks |

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| | B1.1.9—Formation of triglycerides and phospholipids by condensation reactions<br>B1.1.10—Difference between saturated, monounsaturated and polyunsaturated fatty acids<br>B1.1.11—Triglycerides in adipose tissues for energy storage and thermal insulation<br>B1.1.12—Formation of phospholipid bilayers as a consequence of the hydrophobic and hydrophilic regions<br>B1.1.13—Ability of non-polar steroids to pass through the phospholipid bilayer<br><br>**B1.2 Proteins**<br>• Structure and function of proteins- just 2 paragraphs<br>B1.2.1—Generalized structure of an amino acid<br>B1.2.2—Condensation reactions forming dipeptides and longer chains of amino acids<br>B1.2.3—Dietary requirements for amino acids<br>B1.2.4—Infinite variety of possible peptide chains<br>B1.2.5—Effect of pH and temperature on protein structure<br>B1.2.6—Chemical diversity in the R-groups of amino acids as a basis for the immense diversity in protein form and function<br>B1.2.7—Impact of primary structure on the conformation of proteins<br>B1.2.8—Pleating and coiling of secondary structure of proteins<br>B1.2.9—Dependence of tertiary structure on hydrogen bonds, ionic bonds, disulfide covalent bonds and hydrophobic interactions<br>B1.2.10—Effect of polar and non-polar amino acids on tertiary structure of proteins<br>B1.2.11—Quaternary structure of non-conjugated and conjugated proteins<br>B1.2.12—Relationship of form and function in globular and fibrous proteins<br><br>**C1.1 Enzymes and metabolism**<br>C1.1.1—Enzymes as catalysts<br>C1.1.2—Role of enzymes in metabolism | |

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| | C1.1.3—Anabolic and catabolic reactions<br>C1.1.4—Enzymes as globular proteins with an active site for catalysis<br>C1.1.5—Interactions between substrate and active site to allow induced-fit binding<br>C1.1.6—Role of molecular motion and substrate-active site collisions in enzyme catalysis<br>C1.1.7—Relationships between the structure of the active site, enzyme–substrate specificity and denaturation<br>C1.1.8—Effects of temperature, pH and substrate concentration on the rate of enzyme activity<br>C1.1.9—Measurements in enzyme-catalysed reactions<br>C1.1.10—Effect of enzymes on activation energy<br>C1.1.11—Intracellular and extracellular enzyme-catalysed reactions<br>C1.1.12—Generation of heat energy by the reactions of metabolism<br>C1.1.13—Cyclical and linear pathways in metabolism<br>C1.1.14—Allosteric sites and non-competitive inhibition<br>C1.1.15—Competitive inhibition as a consequence of an inhibitor binding reversibly to an active site<br>C1.1.16—Regulation of metabolic pathways by feedback inhibition<br>C1.1.17—Mechanism-based inhibition as a consequence of chemical changes to the active site caused by the irreversible binding of an inhibitor<br>• Classification and Nomenclature of Enzymes<br><br>**A1.2 Nucleic acids**<br>A1.2.1—DNA as the genetic material of all living organisms<br>A1.2.2—Components of a nucleotide<br>A1.2.3—Sugar–phosphate bonding and the sugar–phosphate “backbone” of DNA and RNA<br>A1.2.4—Bases in each nucleic acid that form the basis of a code<br>A1.2.5—RNA as a polymer formed by condensation of nucleotide monomers | |

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| | A1.2.6—DNA as a double helix made of two antiparallel strands of nucleotides with two strands linked by hydrogen bonding between complementary base pairs<br>A1.2.7—Differences between DNA and RNA<br>A1.2.8—Role of complementary base pairing in allowing genetic information to be replicated and expressed<br>A1.2.9—Diversity of possible DNA base sequences and the limitless capacity of DNA for storing information<br>A1.2.10—Conservation of the genetic code across all life forms as evidence of universal common ancestry<br>A1.2.11—Directionality of RNA and DNA<br>A1.2.12—Purine-to-pyrimidine bonding as a component of DNA helix stability<br>A1.2.12—Purine-to-pyrimidine bonding as a component of DNA helix stability<br>A1.2.14—Evidence from the Hershey–Chase experiment for DNA as the genetic material<br>A1.2.15—Chargaff's data on the relative amounts of pyrimidine and purine bases across diverse life forms<br><br>**C1.2 Cell respiration**<br>• Exchange gases;<br>C1.2.1—ATP as the molecule that distributes energy within cells<br>C1.2.2—Life processes within cells that ATP supplies with energy<br>C1.2.3—Energy transfers during interconversions between ATP and ADP<br><br>C1.2.4—Cell respiration as a system for producing ATP within the cell using energy released from carbon compounds<br>• Fermentation(anaerobic), TCA cycle and electron transport system (aerobic)<br>C1.2.5—Differences between anaerobic and aerobic cell respiration in humans<br>C1.2.6—Variables affecting the rate of cell respiration | |

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| | C1.2.7—Role of NAD as a carrier of hydrogen and oxidation by removal of hydrogen during cell respiration<br>C1.2.8—Conversion of glucose to pyruvate by stepwise reactions in glycolysis with a net yield of ATP and reduced NAD<br>C1.2.9—Conversion of pyruvate to lactate as a means of regenerating NAD in anaerobic cell respiration<br>C1.2.10—Anaerobic cell respiration in yeast and its use in brewing and baking<br>C1.2.11—Oxidation and decarboxylation of pyruvate as a link reaction in aerobic cell respiration<br>C1.2.12—Oxidation and decarboxylation of acetyl groups in the Krebs cycle with a yield of ATP and reduced NAD<br>C1.2.13—Transfer of energy by reduced NAD to the electron transport chain in the mitochondrion<br>C1.2.14—Generation of a proton gradient by flow of electrons along the electron transport chain<br>C1.2.15—Chemiosmosis and the synthesis of ATP in the mitochondrion<br>C1.2.16—Role of oxygen as terminal electron acceptor in aerobic cell respiration<br>C.1.2.17—Differences between lipids and carbohydrates as respiratory substrates<br>• Respiratory quotient.<br>• Amphibolic pathways<br><br>**C1.3 Photosynthesis**<br>• Photosynthesis as a means of Autotrophic nutrition;<br>• Site of photosynthesis take place<br>C1.3.1—Transformation of light energy to chemical energy when carbon compounds are produced in photosynthesis<br>C1.3.2—Conversion of carbon dioxide to glucose in photosynthesis using hydrogen obtained by splitting water<br>C1.3.3—Oxygen as a by-product of photosynthesis in plants, algae and cyanobacteria | |

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| | • Pigments involved in Photosynthesis (Elementary idea), Photochemical and biosynthetic phases of photosynthesis<br>C1.3.4—Separation and identification of photosynthetic pigments by chromatography<br>C1.3.5—Absorption of specific wavelengths of light by photosynthetic pigments<br>C1.3.5—Absorption of specific wavelengths of light by photosynthetic pigments<br>C1.3.7—Techniques for varying concentrations of carbon dioxide, light intensity or temperature experimentally to investigate the effects of limiting factors on the rate of photosynthesis<br>C1.3.8—Carbon dioxide enrichment experiments as a means of predicting future rates of photosynthesis and plant growth<br>C1.3.9—Photosystems as arrays of pigment molecules that can generate and emit excited electrons<br>C1.3.10—Advantages of the structured array of different types of pigment molecules in a photosystem<br>• The electron transport, Water splitting, Cyclic and non-cyclic and photophosphorylation<br>C1.3.11—Generation of oxygen by the photolysis of water in photosystem II<br>C1.3.12—ATP production by chemiosmosis in thylakoids<br>• Chemiosmotic hypothesis<br>C1.3.13—Reduction of NADP by photosystem I<br>C1.3.14—Thylakoids as systems for performing the light-dependent reactions of photosynthesis<br>• The Primary Acceptor of CO2, The Calvin Cycle, Photorespiration , C4 pathways;<br>• Factors affecting photosynthesis.<br>C1.3.15—Carbon fixation by Rubisco<br>C1.3.16—Synthesis of triose phosphate using reduced NADP and ATP<br>C1.3.17—Regeneration of RuBP in the Calvin cycle using ATP | |

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| | C1.3.18—Synthesis of carbohydrates, amino acids and other carbon compounds using the products of the Calvin cycle and mineral nutrients<br>C1.3.19—Interdependence of the light-dependent and light-independent reactions<br><br>**D1.1 DNA replication**<br>Genome and human genome project;<br>• DNA fingerprinting.<br><br>• Search for genetic material and DNA as genetic material;<br>• Structure of DNA and RNA; DNA packaging;<br>• DNA replication, Central dogma<br>D1.1.1—DNA replication as production of exact copies of DNA with identical base sequences<br>D1.1.2—Semi-conservative nature of DNA replication and role of complementary base pairing<br>D1.1.3—Role of helicase and DNA polymerase in DNA replication<br>D1.1.4—Polymerase chain reaction and gel electrophoresis as tools for amplifying and separating DNA<br>D1.1.5—Applications of polymerase chain reaction and gel electrophoresis<br>D1.1.6—Directionality of DNA polymerases<br>D1.1.7—Differences between replication on the leading strand and the lagging strand<br>D1.1.8—Functions of DNA primase, DNA polymerase I, DNA polymerase III and DNA ligase in replication<br>D1.1.9—DNA proofreading<br><br>**D1.2 Protein synthesis**<br>• Transcription, genetic code, translation<br>D1.2.1—Transcription as the synthesis of RNA using a DNA template<br>D1.2.2—Role of hydrogen bonding and complementary base pairing in transcription<br>D1.2.3—Stability of DNA templates | |

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| | D1.2.4—Transcription as a process required for the expression of genes<br>D1.2.5—Translation as the synthesis of polypeptides from mRNA<br>D1.2.6—Roles of mRNA, ribosomes and tRNA in translation<br>D1.2.7—Complementary base pairing between tRNA and mRNA<br>D1.2.8—Features of the genetic code<br>D1.2.9—Using the genetic code expressed as a table of mRNA codons<br>D1.2.10—Stepwise movement of the ribosome along mRNA and linkage of amino acids by peptide bonding to the growing polypeptide chain<br>D1.2.11—Mutations that change protein structure<br>D1.2.12—Directionality of transcription and translation<br>D1.2.13—Initiation of transcription at the promoter<br>D1.2.14—Non-coding sequences in DNA do not code for polypeptides<br>D1.2.15—Post-transcriptional modification in eukaryotic cells<br>D1.2.16—Alternative splicing of exons to produce variants of a protein from a single gene<br>D1.2.17—Initiation of translation<br>D1.2.18—Modification of polypeptides into their functional state<br>D1.2.19—Recycling of amino acids by proteasomes<br>Charging of tRNA and Lac operon in detail<br><br>**D1.3 Mutation and gene editing**<br>D1.3.1—Gene mutations as structural changes to genes at the molecular level<br>D1.3.2—Consequences of base substitutions<br>D1.3.3—Consequences of insertions and deletions<br>D1.3.4—Causes of gene mutation<br>D1.3.5—Randomness in mutation<br>D1.3.6—Consequences of mutation in germ cells and somatic cells<br>D1.3.7—Mutation as a source of genetic variation | |

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| | D1.3.8—Gene knockout as a technique for investigating the function of a gene by changing it to make it inoperative<br>D1.3.9—Use of the CRISPR sequences and the enzyme Cas9 in gene editing<br>D1.3.10—Hypotheses to account for conserved or highly conserved sequences in genes<br><br>**D2.2 Gene expression**<br><br>D2.2.1—Gene expression as the mechanism by which information in genes has effects on the phenotype<br>D2.2.2—Regulation of transcription by proteins that bind to specific base sequences in DNA<br>D2.2.3—Control of the degradation of mRNA as a means of regulating translation<br>D2.2.4—Epigenesis as the development of patterns of differentiation in the cells of a multicellular organism<br>D2.2.5—Differences between the genome, transcriptome and proteome of individual cells<br>D2.2.6—Methylation of the promoter and histones in nucleosomes as examples of epigenetic tags<br>D2.2.7—Epigenetic inheritance through heritable changes to gene expression<br>D2.2.8—Examples of environmental effects on gene expression in cells and organisms<br>D2.2.9—Consequences of removal of most but not all epigenetic tags from the ovum and sperm<br>D2.2.10—Monozygotic twin studies<br>D2.2.11—External factors impacting the pattern of gene expression<br><br>**Unit 2: CELLS – 60 hours**<br><br>**A2.1 Origins of cells**<br>A2.1.1—Conditions on early Earth and the pre-biotic formation of carbon compounds<br>A2.1.2—Cells as the smallest units of self-sustaining life<br>A2.1.3—Challenge of explaining the spontaneous origin of cells | |

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| | A2.1.4—Evidence for the origin of carbon compounds<br>A2.1.5—Spontaneous formation of vesicles by coalescence of fatty acids into spherical bilayers<br>A2.1.6—RNA as a presumed first genetic material<br>A2.1.7—Evidence for a last universal common ancestor<br>A2.1.8—Approaches used to estimate dates of the first living cells and the last universal common ancestor<br>A2.1.9—Evidence for the evolution of the last universal common ancestor in the vicinity of hydrothermal vents<br><br>**A2.2 Cell structure**<br>A2.2.1—Cells as the basic structural unit of all living organisms<br>A2.2.2—Microscopy skills<br>A2.2.3—Developments in microscopy<br>A2.2.4—Structures common to cells in all living organisms<br>A2.2.5—Prokaryote cell structure<br>A2.2.6—Eukaryote cell structure<br>A2.2.7—Processes of life in unicellular organisms<br>A2.2.8—Differences in eukaryotic cell structure between animals, fungi and plants<br>A2.2.9—Atypical cell structure in eukaryotes<br>A2.2.10—Cell types and cell structures viewed in light and electron micrographs<br>A2.2.11—Drawing and annotation based on electron micrographs<br>A2.2.12—Origin of eukaryotic cells by endosymbiosis<br>A2.2.13—Cell differentiation as the process for developing specialized tissues in multicellular organisms<br>A2.2.14—Evolution of multicellularity<br><br>**A2.3 Viruses**<br>A2.3.1—Structural features common to viruses<br>A2.3.2—Diversity of structure in viruses<br>A2.3.3—Lytic cycle of a virus<br>A.2.3.4—Lysogenic cycle of a virus<br>A2.3.5—Evidence for several origins of viruses from other organisms<br>A2.3.6—Rapid evolution in viruses | |

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| | **B2.2 Organelles and compartmentalization**<br>• Cell envelope, cell membrane, cell wall;<br>• Cell organelles-structure and function;<br>• Endomembrane system-endoplasmic reticulum, Golgi bodies, lysosomes, vacuoles;<br>• Mitochondria, ribosomes, plastids, microbodies;<br>• Cytoskeleton, cilia, flagella, centrioles (ultrastructure and function);<br>• Nucleus-nuclear membrane, chromatin, nucleolus.<br><br>B2.2.1 Organelles as discrete subunits of cells that are adapted to perform specific functions<br>B2.2.2 Advantage of the separation of the nucleus and cytoplasm into separate compartments<br>B2.2.3 Advantages of compartmentalization in the cytoplasm of cells<br>B2.2.4 AHL Adaptations of the mitochondrion for production of ATP by aerobic cell respiration<br>B2.2.5 AHL Adaptations of the chloroplast for photosynthesis<br>B2.2.6 AHL Functional benefits of the double membrane of the nucleus<br>B2.2.7 AHL Structure and function of free ribosomes and of the rough endoplasmic reticulum<br>B2.2.8 AHL Structure and function of the Golgi apparatus<br>B2.2.9 AHL Structure and function of vesicles in cells<br>B2.3 Cell Specialization<br>B2.3.1 Production of unspecialized cells following fertilization and their development into specialized cells by differentiation<br>B2.3.2 Properties of stem cells<br>B2.3.3 Location and function of stem cell niches in adult humans<br>B2.3.4 Differences between totipotent, pluripotent and multipotent stem cells<br>B2.3.5 Cell size as an aspect of specialization<br>B2.3.6 Surface area-to-volume ratios and constraints on cell size | |

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| | B2.3.7 AHL Adaptations to increase surface area-to-volume ratios of cells<br>B2.3.8 AHL Adaptations of type I and type II pneumocytes in alveoli<br>B2.3.9 AHL Adaptations of cardiac muscle cells and striated muscle fibres<br>B2.3.10 AHL Adaptations of sperm and egg cells<br><br>**B2.1 Membranes and membrane transport**<br>B2.1.1—Lipid bilayers as the basis of cell membranes<br>B2.1.2—Lipid bilayers as barriers<br>B2.1.3—Simple diffusion across membranes<br>B2.1.4—Integral and peripheral proteins in membranes<br>B2.1.5—Movement of water molecules across membranes by osmosis and the role of aquaporins<br>B2.1.6—Channel proteins for facilitated diffusion<br>B2.1.7—Pump proteins for active transport<br>B2.1.8—Selectivity in membrane permeability<br>B.2.1.9—Structure and function of glycoproteins and glycolipids<br>B2.1.10—Fluid mosaic model of membrane structure<br>B2.1.11—Relationships between fatty acid composition of lipid bilayers and their fluidity<br>B2.1.12—Cholesterol and membrane fluidity in animal cells<br>B2.1.13—Membrane fluidity and the fusion and formation of vesicles<br>B2.1.14—Gated ion channels in neurons<br>B2.1.15—Sodium–potassium pumps as an example of exchange transporters<br>B2.1.16—Sodium-dependent glucose cotransporters as an example of indirect active transport<br>B2.1.17—Adhesion of cells to form tissues<br><br>D2.3.1—Solvation with water as the solvent<br>• Movement of water, gases and nutrients<br>D2.3.2—Water movement from less concentrated to more concentrated solutions<br>D2.3.3—Water movement by osmosis into or out of cells | |

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| | D2.3.4—Changes due to water movement in plant tissue bathed in hypotonic and those bathed in hypertonic solutions<br>D2.3.5—Effects of water movement on cells that lack a cell wall<br>D2.3.6—Effects of water movement on cells with a cell wall water relations, imbibition<br>D2.3.7—Medical applications of isotonic solutions<br>D2.3.8—Water potential as the potential energy of water per unit volume<br>D2.3.9—Movement of water from higher to lower water potential<br>D2.3.10—Contributions of solute potential and pressure potential to the water potential of cells with walls<br>D2.3.11—Water potential and water movements in plant tissue<br><br>**C2.1 Chemical signalling**<br>C2.1.1—Receptors as proteins with binding sites for specific signalling chemicals<br>C.2.1.2—Cell signalling by bacteria in quorum sensing<br>C2.1.3—Hormones, neurotransmitters, cytokines and calcium ions as examples of functional categories of signalling chemicals in animals<br>C2.1.4—Chemical diversity of hormones and neurotransmitters<br>C2.1.5—Localized and distant effects of signalling molecules<br>C2.1.6—Differences between transmembrane receptors in a plasma membrane and intracellular receptors in the cytoplasm or nucleus<br>C2.1.7—Initiation of signal transduction pathways by receptors<br>C2.1.8—Transmembrane receptors for neurotransmitters and changes to membrane potential<br>C2.1.9—Transmembrane receptors that activate G proteins<br>C2.1.10—Mechanism of action of epinephrine (adrenaline) receptors | |

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| | C2.1.11—Transmembrane receptors with tyrosine kinase activity<br>C2.1.12—Intracellular receptors that affect gene expression<br>C2.1.13—Effects of the hormones oestradiol and progesterone on target cells<br>C2.1.14—Regulation of cell signalling pathways by positive and negative feedback<br><br>**C2.2 Neural signalling**<br><br>C2.2.1—Neurons as cells within the nervous system that carry electrical impulses<br><br>C2.2.2—Generation of the resting potential by pumping to establish and maintain concentration gradients of sodium and potassium ions<br>C2.2.3—Nerve impulses as action potentials that are propagated along nerve fibres<br>C2.2.4—Variation in the speed of nerve impulses<br>C2.2.5—Synapses as junctions between neurons and between neurons and effector cells<br>C2.2.6—Release of neurotransmitters from a presynaptic membrane<br>C2.2.7—Generation of an excitatory postsynaptic potential<br>C2.2.8—Depolarization and repolarization during action potentials<br>C2.2.9—Propagation of an action potential along a nerve fibre/axon as a result of local currents<br>C2.2.10—Oscilloscope traces showing resting potentials and action potentials<br>C2.2.11—Saltatory conduction in myelinated fibres to achieve faster impulses<br>C2.2.12—Effects of exogenous chemicals on synaptic transmission<br>C2.2.13—Inhibitory neurotransmitters and generation of inhibitory postsynaptic potentials<br>C2.2.14—Summation of the effects of excitatory and inhibitory neurotransmitters in a postsynaptic neuron | |

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| | C2.2.15—Perception of pain by neurons with free nerve endings in the skin<br>C2.2.16—Consciousness as a property that emerges from the interaction of individual neurons in the brain<br><br>**D2.1 Cell and nuclear division**<br>D2.1.1—Generation of new cells in living organisms by cell division<br>D2.1.2—Cytokinesis as splitting of cytoplasm in a parent cell between daughter cells<br>D2.1.3—Equal and unequal cytokinesis<br>D2.1.4—Roles of mitosis and meiosis in eukaryotes<br>D2.1.5—DNA replication as a prerequisite for both mitosis and meiosis<br>D2.1.6—Condensation and movement of chromosomes as shared features of mitosis and meiosis<br>D2.1.7—Phases of mitosis<br>D2.1.8—Identification of phases of mitosis<br>D2.1.9—Meiosis as a reduction division<br>D2.1.10—Down syndrome and non-disjunction<br>D2.1.11—Meiosis as a source of variation<br>D2.1.12—Cell proliferation for growth, cell replacement and tissue repair<br>D2.1.13—Phases of the cell cycle<br>D2.1.14—Cell growth during interphase<br>D2.1.15—Control of the cell cycle using cyclins<br>D2.1.16—Consequences of mutations in genes that control the cell cycle<br>D2.1.17—Differences between tumours in rates of cell division and growth and in the capacity for metastasis and invasion of neighbouring tissue<br>**Unit 3: Organisms**<br>A3.1 Diversity of organisms<br>A3.1.1—Variation between organisms as a defining feature of life<br>A3.1.2—Species as groups of organisms with shared traits<br>A3.1.3—Binomial system for naming organisms<br>A3.1.4—Biological species concept | |

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| | 3.1.5—Difficulties distinguishing between populations and species due to divergence of no ninterbreeding populations during speciation<br>A3.1.6—Diversity in chromosome numbers of plant and animal species<br>A3.1.7—Karyotyping and karyograms<br>A3.1.8—Unity and diversity of genomes within species<br>A3.1.9—Diversity of eukaryote genomes<br>A3.1.10—Comparison of genome sizes<br>A3.1.11—Current and potential future uses of whole genome sequencing<br>A3.1.12—Difficulties applying the biological species concept to asexually reproducing species and to bacteria that have horizontal gene transfer<br>A3.1.13—Chromosome number as a shared trait within a species<br>A3.1.14—Engagement with local plant or animal species to develop a dichotomous key<br>A3.1.15—Identification of species from environmental DNA in a habitat using barcodes | |
| **CHEMISTRY** | **Structure 1. Models of the particulate nature of matter**<br><br>**1.1 Topic Introduction to particulate nature of matter**<br>S1.1.1 The composition of matter<br>S1.1.2 States of matter<br>S1.1.3 Kelvin Temperature Scale<br><br>**S1.2 Nuclear Atom**<br>S1.2.1The structure of the atom<br>S1.2.2 Isotopes<br>S1.2.3 Mass Spectrometry (HL)<br><br>**S1.3 Electron configuration**<br>S1.3.1 Emission spectra<br>S1.3.2 Line emission spectrum of hydrogen<br>S1.3.3 Energy level<br>S1.3.4 The quantum mechanical model S1.3.5 electron configuration<br>S1.3.6 Limit of convergence HL<br>S1.3.7 Successive ionization HL<br><br>**Structure 3.1—The periodic table: Classification of elements**<br>Structure 3.1.1—The periodic table<br>Structure 3.1.2- The period number shows the outer energy level that is occupied by electrons | **SL**<br>Paper 1 (1 hour and 30 minutes)<br>Paper 1A—Multiple-choice questions<br>Paper 1B—Data-based questions<br>(Total 55 marks)<br>Weightage:36%<br>Paper 2 (1 hour and 30 minutes)<br>Short-answer and extended-response questions<br>(Total 50 marks)<br>Weightage: 44%<br><br>**HL**<br><br>Paper 1 (2 hours)<br>Paper 1A—Multiple-choice questions<br>Paper 1B—Data-based questions<br>(Total 75 marks)<br>Weightage+36% |

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| | Structure 3.1.3—Periodicity refers to trends<br>Structure 3.1.4—Trends in properties of elements<br><br>1. Position of hydrogen in the periodic table<br>2. Physical and chemical properties of hydrogen<br><br>Structure 3.1.5—Metallic and non-metallic properties<br>Structure 3.1.6—The oxidation state<br>1. Concept of oxidation and reduction based on electron transfer and oxidation number<br>2. Balancing redox reactions using oxidation number and ion-electron method<br>3. Applications of redox reactions in electrochemistry and metallurgy<br>Structure 3.1.7—Discontinuities occur in the trend of increasing first ionization(HL)<br>Structure 3.1.8—Transition elements(HL)<br>Structure 3.1.9—The formation of variable oxidation states(HL)<br>Structure 3.1.10—Transition element complexes(HL)<br><br>**Structure 1.4—Counting particles by mass: The mole**<br>Structure 1.4.1—The mole<br>Structure 1.4.2—Masses of atoms<br>Structure 1.4.3—Molar mass<br>Structure 1.4.4—The empirical formula<br>Structure 1.4.5—The molar concentration<br>Concentration terms: molarity, molality, mole fraction, parts per million,<br>Structure 1.4.6—Avogadro's law<br><br>**Structure 1.5—Ideal gases**<br>Structure 1.5.1—An ideal gas<br>Structure 1.5.2—Real gases deviate<br>Structure 1.5.3—The molar volume<br>Structure 1.5.4—The relationship between the pressure, volume, temperature and amount of an ideal gas<br><br>**Reactivity 2.1—How much? The amount of chemical change**<br><br>Reactivity 2.1.1—Chemical equations show the ratio of reactants and products in a reaction<br>Reactivity 2.1.2—The mole ratio of an equation<br>Reactivity 2.1.3—The limiting reactant determines the theoretical yield.<br>Green chemistry – eco-friendly reactions, atom economy<br><br>**Structure 2.1—The ionic model**<br><br>Structure 2.1.1—Ions<br>Structure 2.1.2—The ionic bond<br>Structure 2.1.3—Ionic compounds<br>Structure 2.2—The covalent Model<br>Structure 2.2.1—A covalent bond<br>Structure 2.2.2—Single, double and triple bonds<br>Structure 2.2.3—A coordination bond<br>Structure 2.2.4—The valence shell electron pair repulsion (VSEPR)<br>Structure 2.2.5—Bond polarity | Paper 2 (2 hours and 30 minutes)<br>Short-answer and extended-response questions<br>(Total 90 marks)<br>Weightage= 44% |

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| | Structure 2.2.6—Molecular polarity<br>Structure 2.2.7—Carbon and silicon<br>Structure 2.2.8—IMF<br>Structure 2.2.9— The properties of covalent substances<br>Structure 2.2.10—Chromatography<br>Structure 2.3—The metallic Model<br>Structure 2.3.1—A metallic bond<br>Structure 2.3.2—The strength of a metallic bond<br>1. Types of solids – molecular, ionic, metallic, covalent<br>2. Unit cell, packing efficiency, coordination number<br>3. Crystal defects – Schottky and Frenkel defects<br>Types of solutions: solid-liquid, liquid-liquid, gas-liquid, etc. Raoult's Law and its application to ideal and non-ideal solutions Colligative properties: Elevation of boiling point, Depression of freezing point, Osmotic pressure, Relative lowering of vapor pressure, Abnormal molar masses and van't Hoff factor<br><br>**Structure 2.4—From models to Materials**<br>Structure 2.4.1— Bonding is best described as a continuum between the ionic, covalent and metallic models, and can be represented by a bonding triangle.<br>Structure 2.4.2— The position of a compound in the bonding triangle<br>Structure 2.4.3— Alloys are mixtures of a metal and other metals or non-metals. They have enhanced properties.<br>Structure 2.4.4— Polymers are large molecules, or macromolecules, made from repeating subunits called monomers.<br>Structure 2.4.5— Addition polymers form by the breaking of a double bond in each monomer.<br><br>**Structure 3.2—Functional groups:** Classification of organic compounds<br><br>Structure 3.2.1—Organic compounds can be represented by different types of formulas<br>Structure 3.2.2—Functional groups give characteristic<br>Structure 3.2.3—A homologous series<br>Structure 3.2.4—Successive members<br>Structure 3.2.5—"IUPAC nomenclature<br>Structure 3.2.6—Structural isomers<br>Reactivity 3.2.9—Functional groups in organic compounds may undergo oxidation<br>Reactivity 3.2.10—Functional groups in organic compounds<br>Reactivity 3.2.11—Reduction of unsaturated compounds<br>Types of drugs: antacids, analgesics, antibiotics - Chemicals in food: preservatives, sweeteners - Soaps and detergents: cleansing action and structure<br><br>**Reactivity 3.3—Electron sharing Reactions**<br><br>Reactivity 3.3.1—A radical is a molecular entity<br>Reactivity 3.3.2—Radicals are produced by homolytic fission<br>Reactivity 3.3.3—Radicals take part in substitution reactions<br>Reactivity 3.4—Electron-pair sharing reactions | |

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| | Reactivity 3.4.1— Nucleophile<br>Reactivity 3.4.2—In a nucleophilic substitution reaction<br>Reactivity 3.4.3—Heterolytic fission Jan<br>Reactivity 3.4.4—An electrophile<br>Reactivity 3.4.5—Alkenes are susceptible to electrophilic addition reaction<br><br>**Reactivity 1. What drives chemical reactions?**<br><br>**Reactivity 1.1—Measuring enthalpy changes**<br><br>Reactivity 1.1.1—Chemical reactions involve a transfer of energy between the system and the surroundings, while total energy is conserved.<br>Reactivity 1.1.2—Reactions are described as endothermic or exothermic<br>Reactivity 1.1.3—The relative stability<br>Reactivity 1.1.4—The standard enthalpy change<br>**Reactivity 1.2—Energy cycles in reactions**<br><br>Reactivity 1.2.1—Bond-breaking absorbs and bond-forming releases energy. Reactivity 1.2.2—Hess's law. First law of thermodynamics – internal energy and enthalpy Heat capacity, specific heat, enthalpy changes in reactions Hess's law and enthalpy of bond dissociation, formation, combustion<br><br>**Reactivity 1.3—Energy from fuels**<br><br>Reactivity 1.3.1—Reactive metals, non-metals and organic compounds undergo combustion<br>Reactivity 1.3.2—Incomplete combustion of organic compounds<br>Reactivity 1.3.3—Fossil fuels<br>Reactivity 1.3.4—Biofuels<br>Reactivity 1.3.5—A fuel cell<br><br>**Reactivity 2.How much, how fast and how far?**<br><br>**Reactivity 2.2—How fast? The rate of chemical change**<br><br>Reactivity 2.2.1—The rate of reaction<br>Reactivity 2.2.2—Species react as a result of collisions<br>Reactivity 2.2.3—Factors that influence rate of reaction<br>Reactivity 2.2.4—Activation energy, Ea<br>Reactivity 2.2.5—Catalysts increase the rate of reaction<br><br>**Reactivity 2.3—How far? The extent of chemical change**<br><br>Reactivity 2.3.1—A state of dynamic equilibrium<br>Reactivity 2.3.2—The equilibrium law<br>Reactivity 2.3.3—The magnitude of the equilibrium constant<br>Reactivity 2.3.4—Le Châtelier's principle<br><br>**Reactivity 3. What are the mechanisms of chemical change?** | |

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| | Reactivity 3.1—Proton transfer Reactions<br>Reactivity 3.1.1—Brønsted–Lowry acid<br>Reactivity 3.1.2— conjugate acid–base pair<br>Reactivity 3.1.3— Amphiprotic and Amphoteric species<br>Reactivity 3.1.4—The pH scale<br>Reactivity 3.1.5—The ion product constant of water<br>Reactivity 3.1.6—Strong and weak acids<br>Reactivity 3.1.7—Acids react with bases in neutralization reactions.<br>Reactivity 3.1.8—pH curves for neutralization reactions | |
| **PHYSICS** | A.1 Kinematics<br>A.2 Forces and momentum<br>A.3 Work, energy and power<br>A.4 Rigid body mechanics<br>B.1 Thermal energy transfers<br>B.3 Gas laws<br>B.5 Current and circuits<br>B.4 Thermodynamics<br>D.1 Gravitational fields<br>D.2 Electric and magnetic fields<br>D.3 Motion in electromagnetic fields<br>D.4 Induction | **SL**<br>Paper 1 (1 hour and 30 minutes)<br>Paper 1A—Multiple-choice questions<br>Paper 1B—Data-based questions<br>Total 45 marks<br><br>Paper 2 (1 hour and 30 minutes)<br>Short-answer and extended-response questions on standard level material only.<br>Total 50 marks<br><br>**HL**<br>Paper 1 (2 hours)<br>Paper 1A—Multiple-choice questions<br>Paper 1B—Data-based questions<br>Total 60 marks<br><br>Paper 2 (2 hour and 30 minutes)<br>Short-answer and extended-response questions on standard level and additional higher level material.<br>Total 90 marks |
| **COMPUTER SCIENCE** | A1 Computer Fundamentals (A1.1-A1.4)<br>A3 Databases (A3.1-A3.4)<br>A4 Machine Learning (A4.1-A4.2)<br>B1 Computational Thinking (B1.1)<br>B2 Programming (B2.1-B2.5)<br>B3 Object Oriented Programming (B3.1-B3.2) | SL<br>Paper –1- 38 marks - 1 Hr<br>Paper –2 - 50 marks – 1 Hr 15 mins<br>HL<br>Paper –1- 56 marks, 1 Hr 25 mins |

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| | | Paper –2 - 60 marks – 1 Hr 25 mins |
| **BUSINESS MANAGEMENT** | **Unit 1: Introduction to business management**<br>1.1 What is a business?<br><br>1.2 Types of business entities<br><br>1.3 Business objectives<br><br>1.4 Stakeholders<br><br>1.5 Growth and evolution<br><br>1.6 Multinational companies (MNCs)<br><br>**Unit 3: Finance and accounts**<br><br>3.1 Introduction to finance<br><br>3.2 Sources of finance<br><br>3.3 Costs and revenues<br><br>**5.5 Break-even analysis (Note: This topic is taught in year 1)**<br><br>3.4 Final accounts<br><br>3.5 Profitability and liquidity ratio analysis<br><br>3.6 Debt/equity ratio analysis (HL only)<br><br>3.7 Cash flow<br><br>3.8 Investment appraisal<br><br>3.9 Budgets (HL only)<br><br>**BM Toolkit**<br>• SWOT<br>• STEEPLE<br>• Business plan<br>• Gantt Chart<br>• Decision tree<br>• Contribution – HL<br><br>**Unit 4: Marketing**<br><br>4.1 Introduction to marketing<br><br>4.2 Marketing planning<br><br>4.3 Sales forecasting (HL only)<br><br>4.4 Market research<br><br>4.5 The seven Ps of the marketing mix<br><br>4.6 International marketing (HL only)<br><br>**BM Toolkit** | Paper 1- 30 marks- 1 hour 30 minutes for everyone HL/SL<br><br>Paper 2 - 50 marks - 1hour 30 minutes for SL and 1 hour 45 minutes for HL |

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| | 1) SWOT<br>2) Ansoff matrix<br>3) STEEPLE<br>4) BCG matrix<br>5) Business plan<br>6) Gantt Chart ( HL only)<br>7) Decision tree<br>8) Descriptive statistics<br>9) Force field analysis (HL only)<br>10) Contribution – HL only<br>11) Porter's generic startegies (HL only)<br>12) Circular business model | |
| **ECONOMICS** | 1.1 What is Economics?<br>1.2 How do economists approach the world?<br><br>**MICROECONOMICS**<br><br>2.1 Demand<br>2.2 Supply<br>2.3 Competitive market equilibrium<br>**2.4 Critique of the maximizing behaviour of consumers and producers (HL only)**<br>2.5 Elasticities of demand<br>2.6 Elasticities of Supply<br>2.7 Role of government in microeconomics<br>2.8 Market failure—externalities and common pool or common access resources<br>2.9 Market failure—public goods<br>**2.10 Market failure—asymmetric information (HL only)**<br><br>**MACROECONOMICS**<br><br>3.1 Measuring economic activity and illustrating its variations<br>3.2 Variations in economic activity—aggregate demand and aggregate supply<br>3.3 Macroeconomic Objectives<br>3.4 Economics of inequality and poverty<br>3.5 Demand management (demand-side policies) — monetary policy<br>3.6 Demand management—fiscal policy<br>3.7 Supply-side policies | Paper 1 - 25 marks - 1 hr 15 min<br>Paper 2 - 40 marks - 1 hr 45 min<br>Paper 3 - 60 marks - 1 hr 45 min |
| **ENGLISH** | **Unit – Readers, Writers and Text**<br>Paper 1<br>• Comic strips<br>• Political cartoons<br>• Opinion columns-Newspaper, Online<br>• Articles- Magazine, travel, blog, cultural<br>• Infographics<br>• Sci articles<br>• Advertisement print<br>• Graffiti<br><br>**Paper 2** | Paper 1 ( Non - Literary texts)<br>Guided Textual analysis<br>HL- 2 Questions<br>SL - One of the 2 questions<br>Maximum marks -40 (HL) 20 (SL)<br>SL – 1 Hr 15 Mins<br>HL – 2 Hr 15 Mins<br><br>Paper 2 - Comparative Analysis ( Literary Texts)<br>Maximum marks –25<br>SL – 1 Hr 45 Mins<br>HL – 1 Hr 45 Mins |

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| | • Comparative Essay | |
| **ESS – SL** | **Topic 1: Foundation**<br>1.1 Perspectives<br>1.2 Systems<br>1.3 Sustainability<br><br>**Topic 2: Ecology)**<br>2.1 Individuals, populations, communities, and ecosystems.<br>2.2 Energy and biomass in ecosystems.<br>2.3 Biogeochemical cycles<br>2.4 Climate and biomes<br>2.5. Zonation, succession and change in ecosystems<br><br>**Topic 3: Biodiversity and conservation**<br>3.1 Biodiversity and evolution.<br>3.2 Human impact on biodiversity.<br>3.3 Conservation and regeneration.<br><br>**Topic 4: Water**<br>4.1 Water systems<br>4.2 Water access, use and security<br>4.3 Aquatic food production systems<br>4.4 Water pollution<br><br>**Topic 5: Land**<br>5.1 Soil<br>5.2 Agriculture and food<br><br>**Topic 6: Atmosphere and climate change.**<br>6.1 Introduction to the atmosphere.<br>6.2 Climate change—causes and impacts.<br>6.3 Climate change—mitigation and adaptation.<br>6.4 Stratospheric ozone<br><br>**Topic 7: Natural resources**<br>7.1 Natural resources—uses and management<br>7.2 Energy sources—uses and management<br>7.3 Solid waste<br><br>**Topic 8: Human populations and urban systems**<br>8.1 Human populations<br>8.2 Urban systems and urban planning<br>8.3 Urban air pollution | **SL:**<br>Paper 1<br>Duration: 1 hour<br>Marks: 35<br><br>Paper 2<br>Duration: 2 hours<br>Marks: 60 |
| **ESS- HL** | **Topic 1: Foundation (SL and HL)**<br>1.1 Perspectives<br>1.2 Systems<br>1.3 Sustainability<br>***HL Lenses(only for HL) - Environmental law, Environmental economics, Environmental ethics***<br><br>**Topic 2: Ecology (SL and HL)** | **HL:**<br>Paper 1<br>Duration: 2 hours<br>Marks: 70<br><br>Paper 2<br>Duration: 2.5 hours<br>Marks: 80 |

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| | 2.1 Individuals, populations, communities, and ecosystems.<br>2.2 Energy and biomass in ecosystems.<br>2.3 Biogeochemical cycles<br>2.4 Climate and biomes<br>2.5. Zonation, succession and change in ecosystems<br><br>**Topic 3: Biodiversity and conservation (SL only)**<br>3.1 Biodiversity and evolution.<br>3.2 Human impact on biodiversity.<br>3.3 Conservation and regeneration.<br><br>**Topic 4: Water (SL only)**<br>4.1 Water systems<br>4.2 Water access, use and security<br>4.3 Aquatic food production systems<br>4.4 Water pollution<br><br>**Topic 5: Land (SL only)**<br>5.1 Soil<br>5.2 Agriculture and food<br><br>**Topic 6: Atmosphere and climate change. (SL only)**<br>6.1 Introduction to the atmosphere.<br>6.2 Climate change—causes and impacts. 6.3 Climate change—mitigation and adaptation.<br>6.4 Stratospheric ozone<br><br>**Topic 7: Natural resources (SL only)**<br>7.1 Natural resources—uses and management<br>7.2 Energy sources—uses and management<br>7.3 Solid waste<br><br>**Topic 8: Human populations and urban systems (SL only)**<br>8.1 Human populations<br>8.2 Urban systems and urban planning<br>8.3 Urban air pollution | |
| **HISTORY** | Paper 1-HL/SL<br>Prescribed Subject 3: The Move to Global War<br><br>Case study 1: Japanese expansion in East Asia 1931-41<br><br>2.1 Case Study 2: Italian expansion (1933– 1940)<br><br>2.2 Case Study 2: German expansion (1933 - 1940)<br><br>Paper 2-HL/SL<br><br>Topic 10: Authoritarian States (20th Century)<br><br>Topic 12: The Cold War: Superpower tensions and rivalries | Paper 1 – 24 marks<br>1 Hr<br><br>Paper 2 – 30 marks<br>1 Hr 30 Mins |
| **MAT AA SL** | Unit 1: Number and Algebra<br>(SL 1.1 to 1.9)<br>Binomial thm: middle term<br>Sequences: Insertion of A.M. and G.M. between two numbers | Paper 1(Non GDC)<br>80 marks: 90 mins<br><br>Paper 2(GDC)<br>80 marks: 90 mins |

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| | Relation between A.M. and G.M<br><br>Unit 2: Functions<br>(SL 2.1 to 2.11)<br>Sets and their representation<br>Union, intersection, and complement of sets<br>Relations, equivalence relation,<br>Functions – one – one and onto, into<br><br>UNIT 3 Geometry and Trignometry<br>3.1 - 3.8<br><br>Unit 4: Statistics & Probability<br>(SL 4.1 to 4.4)<br><br>Unit 5: Calculus (Differential)<br>SL 5.1 to 5.11, 5.9 deleted<br>Limits, continuity and differentiability. | |
| **MAT AA HL** | Unit 1: Number and algebra<br>(SL: 1.1 to 1.9, AHL: 1.10, 1.11, 1.15)<br><br>Counting principles: Circular and repetitive permutations<br>Binomial thm: middle term<br>Sequences: Insertion of A.M. and G.M. between two numbers<br>Relation between A.M. and G.M<br><br>Unit 2: Functions and algebra<br>(SL: 2.1. To 2.11, AHL: 2.12 to 2.16)<br>Sets and their representation<br>Union, intersection, and complement of sets<br>Relations, equivalence relation,<br>Functions – one – one and onto, into<br><br>Unit 3: Geometry and trigonometry<br>SL: 3.1 to 3.7, HL: 3.8 to 3. 11<br>Unit 4: Statistics<br>SL- 4.1 to 4.4<br><br>Unit 5: Calculus (Kinematics deleted)<br>5.1 to 5.8, 5.10 to 5.17<br>Limits, continuity and differentiability. | Paper 1(Non GDC)<br>110 marks: 2 hrs<br><br>Paper 2(GDC)<br>110 marks: 2hrs<br><br>Paper 3(GDC)<br>55 marks: 1hr 15min |
| **MAT AI SL** | Unit 1: Numbers and Algebra SL<br>1.1 to 1.8 (full unit)<br>Unit 2: Functions SL 2.1 to 2.6 (full unit)<br>Unit 3: Geometry and Trigonometry SL 3.1 to 3.6 (full unit)<br>Unit 4: Statistics and Probability SL 4.1 to 4.6 SL 4.10 | Paper 1(with GDC)<br>80 Marks<br>90 Minutes<br><br>Paper 2(with GDC)<br>80 Marks<br>90 Minutes |

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| **MAT AI HL** | Syllabus Guide:<br>Unit 1: Number & Algebra (SL: 1.1-1.8, AL: 1.9-1.11, 1:14, 1:15)<br><br>Unit 2: Functions (Complete unit SL and AL 2.1-2.10)<br><br>Unit 3: Geometry & Trigonometry (SL 3.1-3.6, HL 3.8,3.14, 3.15, 3.16)<br><br>Unit 4: Statistics & Probability (Complete Unit, 4.1-4.19)<br><br>Unit 5: Calculus (SL: 5.1-5.4, 5.6,5.7, HL: 5.9,5.10) | Paper 1(GDC)<br>110 marks: 2 hrs<br><br>Paper 2(GDC)<br>110 marks: 2hrs<br><br>Paper 3(GDC)<br>55 marks: 1hr 15min |
| **PSYCHOLOGY** | • Learning and Cognition<br>  o Classic and operant conditioning<br>  o Schema theory<br>  o Biological factors in cognitive processes<br>  o Cognitive Models<br>  o Cognitive biases in thinking and decision making<br>  o Social Learning theory<br>  o Cultural factors in cognitive processes<br>  o Environmental influences in cognitive processes<br>  o Potential for improving cognitive processes<br>• Human Development<br>  o Brain Development<br>  o Stage theories and continuous models of human development<br>  o Socio Cultural factors in development<br>  o Enculturation of social norms<br>  o Peer Influence<br>  o Role of Childhood experiences<br>  o Theory of Mind<br>  o Attachment<br>• Health and Wellbeing<br>  o Biological Explanations of Mental Health Disorders<br>  o Prevalence of Health Problems<br><br>Content:<br><br>• Biological Approach<br>• Cognitive Approach<br>• Socio-Cultural Approach<br>  o Emic<br>  o Etic<br>  o Enculturation<br>  o Social Learning theory<br>  o Social Identity theory<br><br>Concept:<br><br>• Bias | **Paper 1 (1 hour 30 minutes) Max Marks: 35**<br><br>Integration of the concepts, content and contexts (35 marks)<br><br>Section A: two compulsory short-answer questions from two of the three content areas<br><br>Section B: two compulsory short-answer questions asking students to apply their knowledge of content to an unseen situation, each from one of four contexts<br><br>Section C: students answer one of two concept-based extended response questions, each from a different context<br><br>**Paper 2 (1 hour 30 minutes) Max Marks: 35**<br><br>Applying concepts and content to research contexts (35 marks)<br><br>Section A: four compulsory questions that focus on the class practical's<br><br>Section B: evaluation of an unseen research study with regard to two or more concepts. |

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| | • Causality<br>• Responsibility<br>• Perspective<br>• Change<br>• Measurement<br><br>All the concepts will be covered | |
| **MODERN LANGUAGES**<br><br>**B SL** | **IDENTITIES**<br>Lifestyles<br>Health and well-being<br>Beliefs and values<br>Subcultures<br>Language and identity<br>**EXPERIENCES**<br>Leisure activities<br>Holidays and travel<br>Life stories<br>Rites of passage<br>Customs and traditions<br>Migration<br>**SOCIAL ORGANISATIONS**<br>Social relationships<br>Community<br>Social engagement<br>Education<br>The working world<br>Law and order<br>**HUMAN INGENUITY**<br>Entertainment<br>Communication and media<br>Technology<br><br>**Text Type:**<br>Letter (formal, Informal), Email, Blog, Diary, Article, Interview writing, Speech Writing, Film review | **Paper-2**<br>Listening - 25 Marks (45 mins)<br>Reading –4 0 Marks (1 Hr)<br>**Paper-1**<br>Writing – 30 Marks (1 Hr 15 mins) |
| **MODERN LANGUAGES**<br><br>**AB INITIO** | **Identities**<br>Personal attributes<br>Personal relationships<br>Eating and drinking<br>Physical well-being<br>**Experiences**<br>Daily routine<br>Leisure<br>Holidays<br>Festivals and celebrations<br>**Social Organisation**<br>Neighbourhood<br>Education<br>The workplace<br>Social issues<br>**Text Type:**<br>Letter (formal, Informal), Email, Blog, Diary, Article, Interview, Invitation, Brochure<br>**Grammar** – Adverbes, articles, pronoms, verbes (présent, passé composé, futur simple), prépositions connecteurs. | **Paper-2**<br>Listening - 25 Marks (45 mins)<br>Reading –4 0 Marks (1 Hr)<br>**Paper-1**<br>Writing – 30 Marks (1 Hr) |

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| **ENGLISH B HL** | **Identities**<br>Lifestyles<br>Health and well-being<br>Beliefs and values<br>Language and Identity<br><br>**Experiences**<br>Leisure activities<br>Holidays and travel<br>Life stories<br>Migration<br><br>**Social organization**<br>Social relationships<br>Community<br>Social engagement<br>Education<br>The working world<br><br>**Sharing the planet**<br>The environment<br>Peace and conflict<br>Ethics<br>Urban and rural environment | **Paper-2**<br>Listening - 25 Marks<br>(1 Hr)<br>Reading – 40 Marks<br>(1 Hr)<br>**Paper-1**<br>Writing – 30 Marks<br>(1 Hr 30 Mins)<br><br>Text types will include the following:<br><br>Blog<br>Diary<br>Email<br>Letter<br>News report<br>Brochure<br>Travel guide |